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INFORMATION ANALYSIS

Formal Grammar Theory in Recognition Methods of Unknown Objects

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Abstract—Questions on the formation of contextual grammars that describe both the structural information of an image and the interaction of images in a complex scenario have been considered. The use of a multilevel grammar is proposed, including the task of parsing a sequence of images, as well as the task of parsing objects for various purposes, when the nature of the source is not clear. It is shown that the formation of a grammar that describes both the structural information of an image and the interaction of images is associated with the need to develop an algorithm for recovering the grammar from a given set of dynamic images that represent a training sample. Some basic provisions inherent in structural methods for describing and recognizing a scene are presented.

Keywords: recognition, grammar, syntactic method, evaluation, solution, image, learning

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INTRODUCTION

Computer vision is one of the most popular areas in the field of intelligent systems, including complex recognition algorithms and complex mechanisms of deep machine learning. Computer vision is based on the reproduction of the mechanisms of the complex system of human vision, which allows one to obtain meaningful information from images, videos, and other visual aids, as well as taking the necessary measures or making recommendations based on the received information. Until recently, computer vision worked only within limited capabilities, but due to advances in artificial intelligence and innovations in deep learning, this area has been actively developed in recent years and has largely surpassed humans in performing some tasks related to the detection and recognition of objects [1–3]. One of the driving factors in the development of computer vision is the amount of data that is used to train and improve such systems [4–7].

The problems that arose in solving the problems of image recognition led to the development of various algorithms, one of which is the structural one, which is called the linguistic or syntactic method [4, 5]. Its peculiarity lies in the fact that the a priori descriptions of classes are structural descriptions, that is, formal constructions in which the principle of taking the hierarchy of the structure of an object into account and taking the relationships that exist between individual elements of this hierarchy within the same levels and between them into account is consistently carried out. Structured recognition is used for a wide variety of tasks: character recognition, speech and fingerprint recognition, scene analysis, and chemical and biological structures [6].

In many cases, a posteriori information about the recognized objects is stored in the records of the corresponding signals, that is, electrocardiograms, encephalograms, infrared and acoustic signals, etc. The syntactic recognition method makes it possible to detect the ordering of structures in signals generated by recognizable objects and to use methods of combinatorial regularity of structures to describe objects, which consist in the fact that, operating with a very limited number of atomic (non-derivative) elements and combination rules, one can obtain a significant variety of descriptions of objects with by unrestricted (for example, recurrent) application of combination rules and the results of the corresponding recurrent procedures to the initial elements. The use of combinatorial regularity as a principle for describing the structure of objects ensures economical use of resources [7].

At the present stage of development of intelligent systems, it has become possible to apply the theory of formal grammars for image recognition by describing an object in the form of a hierarchical structure [8]. The theory of formal languages and grammars was developed in the mid-1950s. and was used to describe

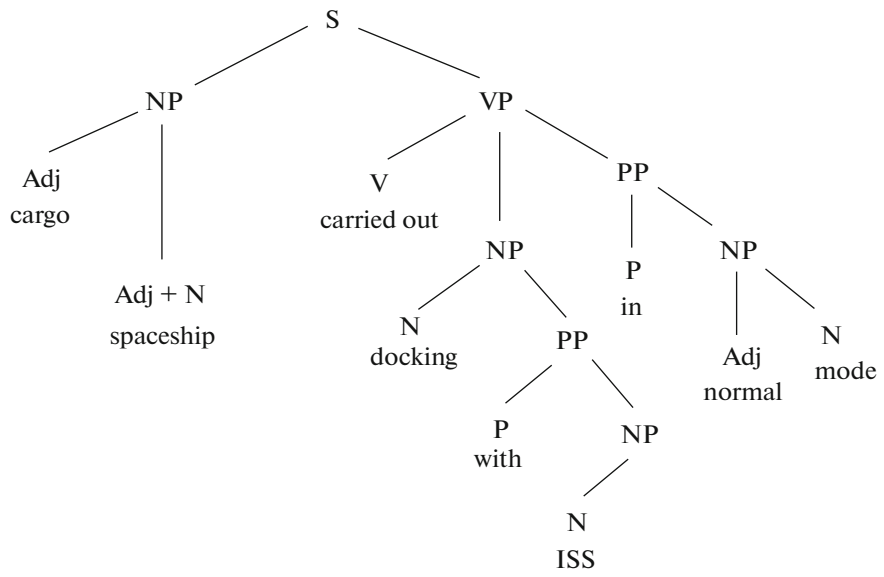

Fig. 1. The syntactic structure of the sentence.

algorithms for processing natural languages, while later it became obvious that a similar mechanism can be transferred to the recognition of images with a complex multi-level structure. In the process of image recognition, elements of an object such as subimages are distinguished, which in turn are subdivided into simpler subimages that are in some relationship with other selected subimages. This allows one to represent information about an object in the form of a hierarchical structure, most often using a graph, in the nodes of which there are subpatterns, and the arcs denote the relationship between the selected subpatterns of the recognized object. Thus, the structural recognition method is extremely similar to the method of analyzing the syntactic structure of natural language sentences [9]. The correspondence between syntagmatics and linear ordering of words in a natural language sentence can be represented as a tree of components. The deep structure of the propositional component of any simple sentence is a construction consisting of a verbal (VP) and a noun (NP) group, which are in some semantic-syntactic relationship. As shown in Fig. 1 the tree of components allows one to represent the syntactic structure of a sentence in the form of a hierarchy composed of nonintersecting units related to each other,

The image recognized by the system can be described in a similar way. Figure 2 illustrates the analogy between parsing a natural language sentence and structured parsing of an image containing geometric shapes.

Within the framework of the presented approach, the solution of the recognition problem is reduced to the use of linguistic constructions containing a certain dictionary of features of the object that is being recognized and a grammar, according to the rules of which the elements of this object are described. After each subimage within the object is identified, its recognition is performed by parsing a sentence describing this object in the form of a tree structure in order to determine whether such a sentence is syntactically correct with respect to the specified grammar. Sentence templates are defined as lined up sequences of selected subimages in various ways, just as sentence phrases are built by concatenating words, and words by concatenating morphemes. The syntactic approach to pattern recognition makes it possible to describe a variety of subpatterns using small sets of simple grammatical rules of formal grammar [6, 9].

THE BASIC CONCEPTS OF FORMAL GRAMMAR

Since the implementation of structural methods of pattern recognition is based on the apparatus of a formal grammar, let us consider some of its basic concepts that are necessary for understanding the essence of the structural approach.

The theory of formal languages as an independent discipline, as a rule, originates from the works of the famous American linguist Noam Chomsky, where he attempted to give an accurate and unique characterization of the structure of natural languages in the 1950s. Its goal was to define the syntax of languages using simple and precise mathematical rules. It was later discovered that the syntax of programming languages can be described using one of Chomsky's grammatical models called free-context grammar [10]. Soon after the advent of modern computers, it became obvious that all forms of information, whether numbers, names, images, or sound waves, can be represented as

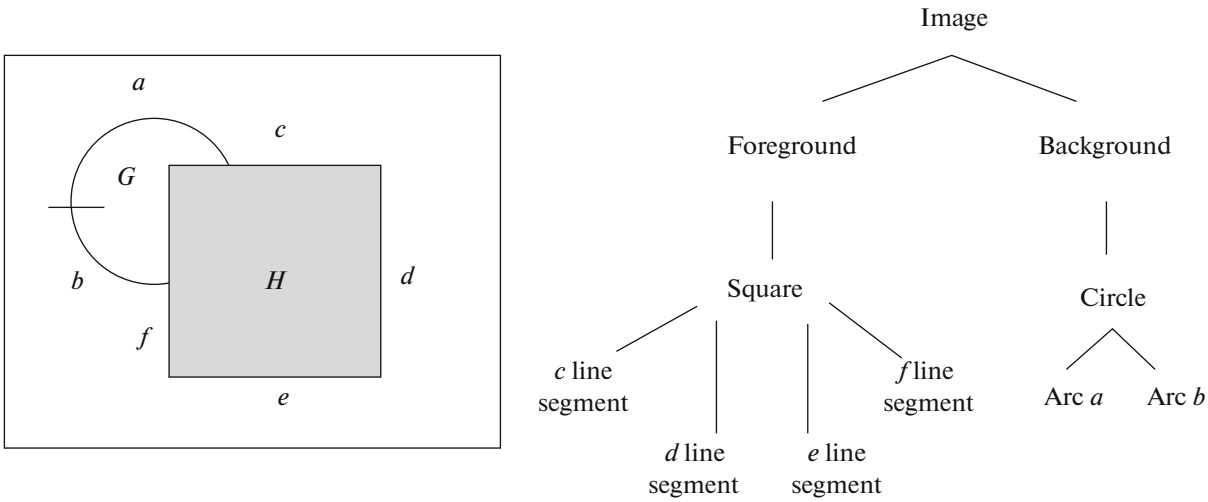

Fig. 2. Parsing an image.

elements written in a string according to certain rules. The theory of formal grammars has taken center stage in applied research in artificial intelligence. Moreover, using the apparatus of formal grammars, it is possible not only to describe the structure of an object, but also to create a parser to determine whether a sentence describing an object is syntactically correct, i.e., whether it belongs to a particular language or is grammatically incorrect. Separate formalisms are used to describe such recognition models in the theory of formal languages, which are known as the theory of automata [11].

In formal language theory, grammar G is a set of rules for generating strings in a formal language. The rules describe how to form strings from the alphabet of the language that are valid in accordance with the syntax of the language. The alphabet is a finite nonempty set of characters. Symbols are assumed to be indivisible. A word (string) in the alphabet Σ is a finite sequence of elements. Generative grammar G defined as a production system consisting of an ordered four $G = \langle T, N, I, P\rangle$ where T is a set of terminal symbols of the alphabet Σ, that is, the initial elements of the dictionary, N is a finite set of nonterminal (auxiliary) symbols, I is the starting character of the grammar, and P is a finite set of rules.

The construction of a grammar begins with a start character I. By applying the rules first to I and then recursively to the result of the previous transformation one can generate the correct "sentence" of the formal language defined by the grammar. The grammar is constructed using special rules for substitution of expressions of the form "$x \to at$", which means "replace x with at" Or "substitute x instead at", where x and at are strings containing any terminal or nonterminal symbols. The conversion algorithm stops when the expression no longer contains nonterminal alphabet characters. Thus, in accordance with the rules of grammar, an object is represented by a sentence in this language (Fig. 3). To further describe the generation process, it is necessary to introduce such concepts as direct deducibility, deducibility, and the language generated by grammar [10, 11].

If there are chains x and at, which can be represented as $x = z_1az_2$ and $y = z_1bz_2$, where $and \to b$ are one of the grammar rules G, then the chain at is directly derivable from the chain x in grammar G. In other words, the chain x can be recycled into the chain at in one step by applying one substitution: x comes from at replacing b in place of some occurrence of the chain and. This is denoted as $x \mid = at$ or $x \mid = y$.

If there is a sequence of strings $x_0, x_1,..., x_n$, in which each next chain is directly deducible from the previous one, then the chain x_n is derivable from the chain x_0. A sequence of chains $x_0, x_1,..., x_n$ is called inference x_n of x_0 in grammar G. This means that the chain x_0 processed into chain x_n not necessarily in one step, but by sequential application of several substitutions. It is obvious that direct deducibility is a special case of deducibility. Output that begins with a start character and ends with a string of only terminal characters is called complete output. Naturally, not every output that begins with a start character is complete; possible outputs that begin with a start character that cannot be continued until the full output are dead-end outputs. The inability to continue the output, although it is not complete, that is, it does not end with a chain of terminal symbols, is determined by the fact that the corresponding grammar does not contain a rule whose left side would be contained in the last chain of the given output. We believe that a "good" grammar is not required such that any output in it ends with the correct terminal chain: it is enough for any complete output to give the correct chain. As a result of constructing full deductions from the initial character in the

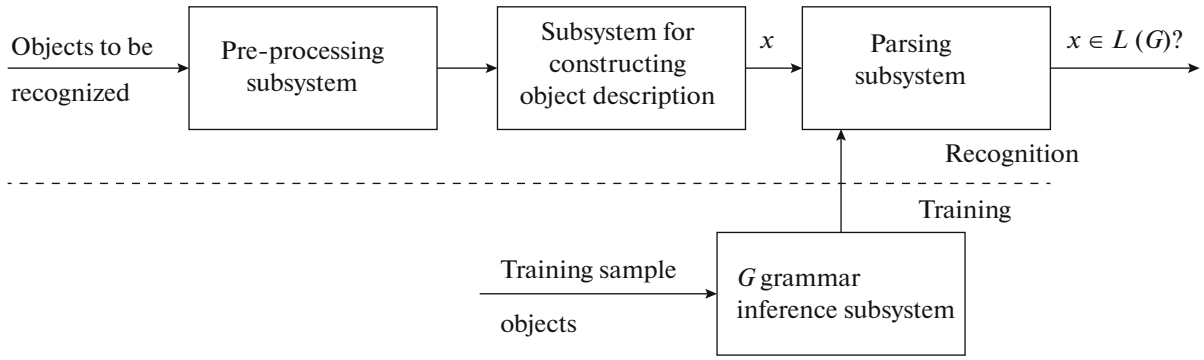

Fig. 3. Object presentation grammar.

grammar a chain consisting of only terminal characters is obtained. This chain is called the language generated by the grammar [9–11].

It is important to consider that a well-formed grammar should not contain useless and inaccessible characters.

Definition. The symbol X is called useful in the grammar $G = \{T, N, I, P\}$ if there is an output of the form $A \Rightarrow *\alpha X\beta \Rightarrow *w$, where $w \in N*$.

We note that X can be either a variable or a terminal and the output string $\alpha X\beta$ is first or last in the output.

Definition. If the symbol X is not useful, it is called useless. Obviously, eliminating useless characters does not change the grammar-generated language; thus, all such characters can be removed.

Definition. Symbol X is called generating in the grammar $G = \{T, N, I, P\}$ if there is an output of the form $X \Rightarrow *w$ where $w \in N*$.

We note that each terminal is a parent symbol, since it logs itself out in 0 steps.

Definition. X is called reachable in grammar $G = \{T, N, I, P\}$ if there is an output of the form $A \Rightarrow *\alpha X\beta$ for some α, β.

A useful symbol, as follows from the definition, is both generative and attainable. If one remove from the grammar first nongenerating and then unreachable symbols, then only useful ones will remain.

Example. We consider grammar $G = \langle\{A, B, C, a, b\}, \{a, b\}, I, P\rangle$ with rules $A \to BC$; $A \to a$; $B \to b$.

All characters except C, are generating, since the terminals a and b give birth to themselves; A generates a and B generates b. Deleting B leads to the deletion of the rule $A \to BC$, i.e., to the grammar $G = \langle\{A, B, a, b\}, \{a, b\}, P, A\rangle$ with rules $A \to a$; $B \to b$.

According to the hierarchy of N. Chomsky [12], formal grammars are divided into four types, where each subsequent type is a subset of the previous one:

Type zero grammar (recursively enumerable, unbounded). Such a grammar includes formal grammars of all types and admits any inference rules. The rules can be written as $a \to b$ where no restrictions are imposed on lines *and b*. Parsing a sentence of such a language can be done using a Turing machine;

Type 1 grammar (context sensitive grammar). For context-sensitive grammars, the left side of the rule can indicate the context in which a particular rule applies; thus, the same nonterminal element can be transformed differently depending on the context in which it appears. Context sensitive grammars have rules of the form: $xYz \to xyz$, where the substitution $Y \to y$ is only valid in context x, z. Parsing a sentence of such a language can be done with a linearly bounded machine;

Grammar of the second type (context-free grammar). A grammar like this is generated by context-free languages, all of whose rules take the form $A \to b$, where $A \in T$, $\beta \in (T \cup N)^*$. Rule substitutions within a grammar are context independent. Parsing a sentence of a given language can be performed using a machine with a store (stack) memory;

grammar of the third type (regular). The rules are written as $A \to a$, $A \to a$, B, where AND, $IN \in N$, $a \in T^*$. Parsing a sentence of the third type of language can be done using a state machine.

REALIZATION OF RECOGNITION PROCESS OBJECTS BASED ON STRUCTURAL METHODS

To recognize an unknown object on the basis of structural methods, it is first of all necessary to find its nonderivative elements and the relationships between them, and then, using parsing (grammatical analysis), establish whether the description of the image is consistent with the grammar that, by assumption, could generate it [13, fourteen].

For the formation of the corresponding grammar, one can use either a priori information about the recognized objects, or the results of studying a finite sample set of "typical" objects in a sense. In the first case, they consider specifying a grammar based on heuristic considerations; in the second, they consider the derivation of grammar.

Let us assume that we are dealing with two classes of objects (Ω_1 and Ω_2), and objects belonging to these classes can be described using features belonging to a certain finite set. These signs can be considered as the main symbols (terminal) and are denoted by T. Each object can be thought of as a string or a sentence, as it is composed of elements of the main vocabulary.

Let there also be a grammar G such that the language generated by it consists of sentences (objects) that belong exclusively to one of the classes, for example, the class Ω_1. This grammar can be used to classify objects, since the presented unknown object can be attributed to the class $\Omega_1$if it is a language sentence L (G). Otherwise, the object is assigned to the class Ω_2. In this case, the object falls into the class Ω_2 only because it does not belong to the class Ω_1: if it turns out that the object is not a grammatically correct sentence in the sense of grammar G, then it is assumed that it must belong to the class Ω_2. In fact, the object may not belong to the class Ω_2. It can also represent a noisy or distorted chain, which in some cases it is preferable to simply remove it from consideration. To do this, one need to have grammars G_1 and G_2 that generate languages L (G_1) and L (G_2), respectively. The object is credited to the class in the language of which it turns out to be a grammatically correct sentence. If the latter fails, then it is obvious that the given object does not belong to either of the two given classes and, therefore, another grammar is required G_3, etc. In case m classes are considered m grammars and related languages L (G_i), $i = 1, 2, ..., m$. The object to be recognized belongs to the class Ω_2 if and only if it is a grammatically correct sentence of the language L (G_i). If an object turns out to be a grammatically correct sentence of more than one language, then the decision regarding its belonging is made in exactly the same way as when using other recognition methods, when it turns out that the results of the implementation of the recognition procedure do not allow the object to be definitely assigned to one of the specified classes.

For the practical use of the structural method, it is required to solve the following main problems:

(1) construction of an adequate description of the recognized objects;

(2) choice of grammar;

(3) the implementation of the recognition process by parsing procedures;

(4) the use of learning procedures for the derivation of grammars;

(5) application of procedures from other recognition methods within the framework of the structural approach (for example, statistical to account for noise and random distortions, cluster analysis, etc.).

Let us consider the basic techniques used in the practical application of structural recognition methods.

Methods of Mathematical Morphology

In the process of image recognition based on structural methods, one important stage is preliminary processing of the analyzed image. At this stage, the object presented for recognition is subjected to encoding and filtering, restoration and image quality improvement. Such filtering methods are considered within the framework of mathematical morphology, which provides an effective approach to the analysis of digital images [15, 16]. Within the framework of morphological filtering, it is not the analytical properties of signals that are used, but their geometric characteristics. The object is encoded or approximated in such a way that it would be convenient to work with it further. For example, a black and white image can be encoded using a grid (or matrix) of ones and zeros. To improve the efficiency of processing in subsequent stages, some kind of "data compression" is often used. Filtering and restoration techniques are then used to remove the distortion to improve image quality. It is assumed that images of sufficiently good quality are reproduced at the end of the preprocessing.

Selecting Nonderived Elements

The first step in constructing a structural description of a recognition object is to define a set of nonderived elements with which one could describe the object in question. This choice depends on the nature of the object, the type of initial data, the area of application, and the method of practical implementation of the recognition system. Currently, there is no universal solution to the problem of choosing nonderived elements. Typically, when selecting nonderived elements, the following two main criteria are used:

(1) Nonderivative elements should serve as the main elements of the image in order to ensure that a compact and adequate description of the initial data is obtained using well-defined structural relationships (for example, connection relationships);

(2) Nonderivative elements should be easy to isolate or recognize using well-known methods of a nonstructural nature, since it is assumed that they are simple and compact images, whose "internal" structural information is not taken into account.

Thus, for speech, it is natural to consider phonemes as a "good" set of nonderivative elements on which the connection relationship is set. Likewise, strokes (dashes) are selected as nonderivative elements to represent handwritten characters.

However, there is no universal element for an image at all, like a phoneme or stroke for a specific object, that is, speech or handwritten symbols. Sometimes, for an adequate description, it is necessary that nonderivative elements contain information that is essential in recognizing objects or phenomena. If, in particular, when solving the problem of object recognition, its size (or shape, or location) is essential, then

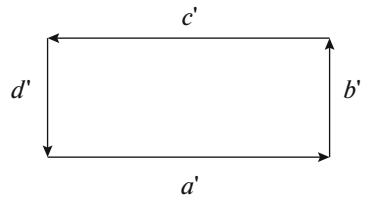

Fig. 4. An example sentence for describing a rectangle.

nonderived elements should carry information about size (or shape, or location) so that objects belonging to different classes turn out to be different regardless of what method is used to parse their corresponding descriptions. This condition, however, often leads to the need to resort to semantic information when describing nonderived elements.

Example. Let us say one need to be able to distinguish rectangles of different sizes from other shapes.

The following set of nonderived elements is selected: a'—0° is a horizontal line segment; b'—90° is a segment of a vertical line; c'—180° is a segment of a horizontal line; and d'—270° is a segment of a vertical line.

The set of all possible rectangles (of various sizes) is given with one sentence: the chain $a'b'c'd'$ (Fig. 4). If it is required to distinguish between rectangles of different sizes such a description is inadequate.

In this case, segments of unit length should be used as nonderived elements. Many rectangles of various sizes can be described using the language:

$$L = \left\{a^n b^m c^n d^m,\ n,\ m = 1, 2, \ldots\right\}.$$

In some cases, the second criterion for selecting nonderivative elements may conflict with the first criterion because the nonderivative elements selected in accordance with this first criterion may not be easily recognizable using currently known methods. However, the second criterion allows the selection of sufficiently complex nonderivative elements, as long as they can be recognized. The complication of nonderived elements makes it possible to simplify structural descriptions, that is, it allows the use of less complex grammars. Finding a compromise solution is essential when building real recognition systems based on the use of structural methods.

Using grammars and languages to describe objects structurally

A grammar can be used to generate sentences that represent an object and to parse sentences to determine whether the structure of those sentences is valid in terms of the corresponding grammar.

Example. We consider grammar $G = \{T, N, I, P\}$, where $T = \{a, b, c, d\}$; $N = \{I, A, B, C, D\}$;

$$P\colon I \to aA\ A \to b\ I \to cC\ C \to d$$

$$I \to bB\ B \to c\ I \to dD\ D \to a.$$

In this case, four sentences can be output (⇒—output character):

$$I \Rightarrow aA \Rightarrow ab;\ I \Rightarrow cC \Rightarrow cd;$$

$$I \Rightarrow bA \Rightarrow bc;\ I \Rightarrow dD \Rightarrow Da.$$

If we consider the oriented segments shown in Fig. 5*c*, then using this grammar one can obtain four descriptions of segments that form a right angle (Fig. 5*b*). Below is a technique that allows one to describe two-dimensional objects using chain grammars. This technique consists in connecting structures only at specific points. One way to implement this requirement is to have only two points allocated in each structure. In our case, the selected points are interpreted as the "head" and "tail" ends of the arrow.

Output Trees

In computer science, a tree is understood as a widely used abstract data type, similar to the hierarchical structure of a tree, with a root value and subtrees, that is, offspring with a parent node, represented as a set of related nodes [17, 18]. Each node of the tree has a certain degree, which characterizes the number node subtrees. A node with a degree zero is called a sheet. The leaves are nodes from which no branches emerge.

The process of outputting a sentence describing a rectangle in the form of a parse tree is shown in Fig. 5. The start character *I* acting as the root of the tree and the leaf nodes read from left to right form the output sentence.

In essence, any hierarchically ordered schema allows one to represent the corresponding object in the form of a tree. Thus, in Fig. 6 ordering consists in grouping the regions of the recognized object; region *b*

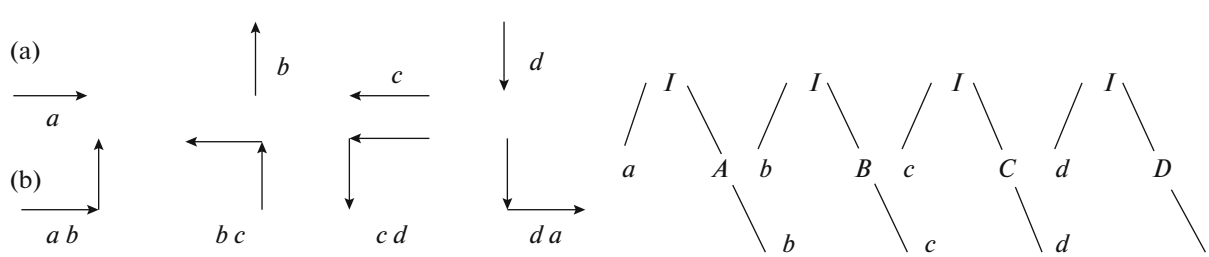

Fig. 5. Sample output using a tree.

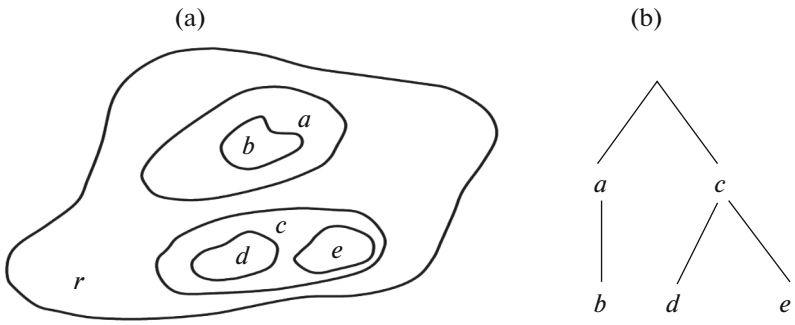

Fig. 6. A schema of ordered objects: a) grouping of areas of the recognized object; b), output tree.

is in region *a*, which in turn is located in region *r*. Figure 6b depicts a tree structure that naturally follows from this sequencing scheme. Each grammar output tree specifies one sentence. Most grammars allow one to obtain a significant, and when introducing probabilistic inference rules, an infinite number of sentences.

In the simplest case, in such trees, two branches extend from each node: one corresponds to the presence of a certain feature in the object, the other to its absence. Such trees do not carry any information about the process of deriving solutions in the above sense; they give a hierarchical representation of which features (in the case of structural recognition, what nonderivative elements and relations) characterize the object under study. However, it is possible to form a grammar describing the inference process in accordance with the structure of a decision tree (usually, only a tree is built and used in this method).

Recognition procedures associated with the use of decision trees belong to the group of structural methods, although in the strict sense decision trees are a tool for the hierarchical separation method [18], which is used in cases where the decision tree "contains" many features. In each node of the tree, one feature is studied and, depending on the results of this study, the next branch of the tree is determined. The classification result is revealed on the lower tier of the tree. Such a recognition scheme is very convenient for taking a priori information about objects into account, but there are no optimal training procedures for it.

IMPLEMENTATION OF THE OBJECT RECOGNITION PROCESS USING GRAMMAR

The importance of formal grammars for object recognition lies, in particular, in the fact that they allow one to see syntactic correctness or incorrectness in relation to a certain grammar of the representation of the image under study using given nonderivative elements and relations. Similar elements and relationships can be obtained by the procedure in [19], which uses two main types of parsing: top-down parsing and bottom-up parsing. Top-down parsing consists in successive attempts, using the rules of the appropriate grammar, to obtain a given terminal sentence based on the initial character. When parsing from bottom to top, it is necessary to restore the inference tree, starting with the elements of the main dictionary and applying the inverted substitution rules. This procedure starts with a specific sentence and ends with a start character. Thus, a nonderivative element *a* included in the proposal *ab*, can be obtained as a result of the transition $D \to a$, and as a result of the transition $I \to aA$; element *D* arises from the substitution $I \to dD$; thus, this conclusion does not result in the proposal *ab*. Since the substitutions $I \to aA$ and $A \to b$ really generate *ab*, this sentence is considered grammatically correct; *ace* and *ad* deviate. The described schemes of grammatical parsing are, in principle, ineffective, since they provide a full enumeration when applying substitution rules. It is often possible to avoid replaying the entire sequence of substitution rules by checking the validity of the intermediate results, thereby determining whether the given sequence of rules ensures successful parsing.

Improving the efficiency of parsing can be achieved by using syntactic rules that establish admissible relationships or prohibit some specific relationships between objects.

Example. We consider grammar $G = \{T, N, I, P\}$, where $T = \{a_1, a_2\}$, $N = \{I, O_1, O_2\}$, P: $I \to$ above (a_1, O_2), $A_2 \to$ above (O_1, a_1), $O_1 \to$ left (a_2, a_1), which is capable of generating squares (involuntary elements, which are horizontal and vertical line segments of fixed length); relationship above (*x*, *at*) and left (*x*, *at*) are read accordingly "*x* is located above *at*" and "*x* is located to the left of *at*." While *x* is necessarily located directly above the segment *at* and part of the segment *x,* in the second case it is located directly to the left of the segment *at*.

The consistent application of the grammar rules leads to structures, whose construction begins with a horizontal segment, then another horizontal segment is introduced, located above the first, after which two vertical segments are placed between them (see Fig. 2). A variety of designs is provided by imposing con-

straints on positional relationships above them and to the left.

In the same example, we will show the order of execution of syntactically oriented parsing. Let it be required to establish whether some construction belongs to the class objects generated by this grammar. When parsing from bottom to top, first of all, an attempt is made to find the object O_1 by going into a nonderived element object a_2 to the left of the nonderived element a_2. In the positive case, parsing continues, in the negative case, the object is rejected. Since the bottom-up parsing procedure begins with a terminal sentence, at the first step only substitution rules are considered that lead exclusively to nonderived elements. The next step of parsing is to obtain the object O_2 consisting of object O_1 located above the nonderived element a_1. If the result is positive, an attempt is made to display the initial character I looking for a nonderivative element a_1 located above the object O_2. If the result is positive, the image is considered acceptable, if it is negative, it is rejected.

Inference of Grammars

Let us consider, using a specific example, a method of obtaining a grammar suitable for describing objects within the framework of a structural approach to recognition using a learning procedure. In essence, the desired grammar is formed on the basis of studying a finite sample of sentences of a certain language, while sometimes a finite sample of sentences not related to this language is also taken into account. The elements of the first sample must be structurally complete in the sense that each rule of the desired grammar is used to generate at least one sentence of the sample.

Figure 6 shows a diagram of the output of chain grammars. The problem is that the set of sample chains $\{x_i\}$ is processed using the adaptive training algorithm (*output algorithm*). At the output of the block, as a result of the learning process, the grammar G is reproduced consistent with the given chains. The procedure for solving this problem in the general case is still unknown; there are, however, many proprietary algorithms. A typical approach can be considered the following: first, substitution rules are constructed that generate exactly those chains that were specified; then, by splicing the elements of the main vocabulary, a simpler recursive grammar that provides the generation of an infinite number of strings. The procedure consists of three stages:

(1) formation of substitution rules;

(2) transformation of substitution rules into grammar;

(3) simplification of grammar.

Let us demonstrate its derivation using the example of a two-dimensional grammar, since, as indicated above, it is this type of grammar that is currently of the greatest interest from an applied point of view. It is assumed that a finite set of relationships, a finite list of elements of the main vocabulary, i.e., nonderived elements, and a set of object descriptions $\{S_1,..., S_n\}$ occur. Each description contains a list of nonderived elements included in it, each of which can be supplied with some additional information. The challenge is to define a "good" set of grammar rules written in terms of a given relationship and consistent with a given set of objects. Thus, a grammar should be formed so that when any of the given objects are introduced into the grammar analyzer and analyzed in accordance with the resulting grammar, one or more structural descriptions of the object are reproduced at the output of the analyzer.

The first step of the inference process is performed separately and independently for each "input" object. As a result of this, a structure should emerge that is set by the selected system of relationships, in this case, predicates that characterize both the properties of nonderivative elements and the relationships between them. At the same step, for each element of objects, the validity of their predicates is determined. If this is positive, a new element is formed and information about the corresponding implicants and the truth of the predicate is fixed in the form of a certain grammar rule. The next loop is then executed, in which the predicates are checked on the elements built from new elements and elements of the input objects; the procedure is repeated until the construction of new elements is impossible. In each successive cycle, naturally, only those descriptions are checked that contain at least one element formed in the previous cycle [20].

At the end of the first step, it is determined which of the elements include all nonderived elements of the corresponding input object as their implicants. All unverified descriptions and descriptions included as an implicant in verified descriptions are excluded from further consideration. Accordingly, all grammar rules that were not used in the construction of the left descriptions are also excluded. The descriptions obtained in this way are a kind of tree of structural descriptions, which are first subjected to "cleaning" (in particular, trees are identified that are equivalent as a result of replacing specific nonderived elements with their types).

For each input object, one or more grammars are thus constructed that correspond to different structural descriptions obtained for the input object. Only those elements of the auxiliary vocabulary that are found in more than one of the constructed grammars correspond to the highest level of description, that is, the description of the object as a whole.

The second step of the withdrawal process is as follows. A union of grammars obtained in the first step for each of the objects is built. A special transformation procedure is applied to each of these grammars and the best one is selected from the obtained modified grammars according to some criterion. Of course,

preference is given to short grammars, since they really are a single grammar for a given training set, and not just a set of special cases. However, it is advisable that the substitution rules be as specific as possible with regard to the possible types of nonderivative elements and relations. Obviously, the final criterion for choosing a grammar must be based on some combination of these two heuristic techniques.

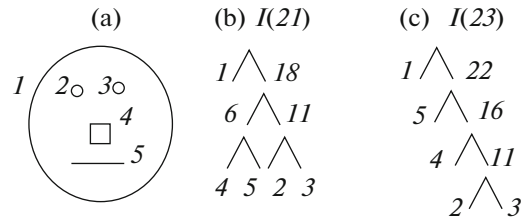

Fig. 7. Representation of predicates using grammars.

Converting Grammar

For effective recognition of unknown objects using a formal grammar, it must be reduced, i.e., redundant rules should be eliminated. For this, a grammar transformation procedure is applied, which includes several main stages:

(1) All redundant occurrences of rules are eliminated if some rule is included in the transformed grammar G repeatedly;

(2) Such a pair of elements of the auxiliary dictionary is found that the identification of these elements (replacing one with another in the entire grammar) will lead to the appearance of redundant occurrences of the rules. If one have different choices for such pairs, one should select the pair that results in the most redundant occurrences. After identifying the rules, one should return to the first stage;

(3) Such a pair is sought N, n containing an element of the auxiliary dictionary N and nonderivative element n that inclusion in grammar G regulations $N \to n$ (if not included) and selective replacement N on the n in grammar Γ lead, after eliminating the multiple occurrences of the rules, to reducing their number. If there are various options for choosing such pairs, one should choose the pair that provides the greatest reduction in the number of rules in the grammar. Then, one should return to the first stage;

(4) A threshold is introduced according to the number of rules identified at stages 1–3: the replacement is performed if the number of occurrences or identified rules is not less than the threshold value. This is done in order to not substantially diminish the separative power of the grammar;

(5) A threshold is introduced for the number of rules that are identical in everything (the exception is the type of a nonderived element): if the number of such rules is greater than or equal to the threshold value, then the corresponding rules are replaced by one containing a nonderived element of an arbitrary type. One should then return to the first stage.

If further simplifications are not possible, then the procedure is terminated. The result is one grammar. After this procedure is performed for each of the grammars generated in the first step, one needs to choose the "best" grammar. As a criterion for the optimality of grammar, one can use, for example, the ratio of the separating force of the grammar to the square of the number of its rules, where the dividing force of a grammar is defined as the sum of the dividing forces of each of its rules, which are defined as the sum of its nonderivative elements and terms that represent a relationship or property.

As an example let us consider the situation where only one input object is presented, as shown in Fig. 7. Here, nonderived elements are of the following types: circle, point, square, and line segment. We suppose the following predicates: left (x, *at*) (executed if the element x located to the left of the element *at*); higher (x y) (executed if the element x is above the element *at*); and inside (x, *at*) (executed if the element x is inside the element *at*). In this case, the input object is represented as a list of five nonderived elements: a point (two), circle, square, and line segment. First, all cases are selected where the elements of the input list satisfy one of the given predicates. Each executed predicate forms a new element. Next, it is determined which of the new elements can be built from the original newly formed ones; this procedure is then repeated with the next built elements and so on until the possibilities for constructing new elements are exhausted. After this, all elements that are not components of some element containing all nonderived elements of the input object are removed. In practice, it takes this form. Nonderived elements are numbered 1–5. We suppose, for example, an element above (4, 5) is found: the corresponding predicate "above (4, 5)" is executed. The executed predicate forms a new element whose components are nonderived elements 4 and 5.

Each of the structural descriptions obtained in this way is directly translated into the grammar (the symbols $G1$, $G2$, $G3$ are used as necessary to denote elements of the auxiliary vocabulary and the elements of the main vocabulary are replaced by a nonderived element of a certain type).

Thus, the two constructed grammars are as follows:

grammar 1) I $\to$ (x, y): $G1$(x), circle (*at*): inside (x, y): $G1 \to$ (x, y): $G2$ (x), a straight line segment (*at*): higher (x, y): $G2 \to$ (x, y): $G3$ (x), square (*at*): higher (x, y); $G3 \to$ (x, y): point (x), point (*at*): left (x, y);

grammar 2) I $\to$ (x, y): $G4$(x), circle (x): inside (x, y); $G4 \to$ (x, y); $G5$ (x), $G6$ (*at*): higher (x, y); $G5 \to$ (x, y): point (x), point (*at*): left (x, y); $G6 \to$ (x, y): square (x), a straight line segment (*at*); higher (x, y).

The structural approach to object recognition does not yet have a coherent and rigorous mathematical theory. The main areas of application of structural recognition at present are the processing and analysis of signals of various natures, two-dimensional and three-dimensional images. The practice of constructing recognition systems indicates that it is advisable to use structural methods in conjunction with other methods of recognizing objects and phenomena. The degree of applicability of structural methods will be determined by the development of their mathematical theory and the presence of neural networks oriented towards it.

We can note the new normal forms of context-free grammars, in which all rules have one of two forms:

$$U \to u;$$
$$U \to TN,$$

where U, N, T are nonterminal symbols and u is the terminal symbol. This form is called the Chomsky normal form.

CONCLUSIONS

To recognize multilevel and poorly structured objects, one need to choose the best grammar. Recognition of an unknown object based on structural methods requires first of all finding its nonderivative elements and relationships between them, and then, using parsing (grammatical analysis), establishing whether the description of the image is consistent with the grammar that, by assumption, could have generated it. To form the appropriate grammar, one can use a priori information about the objects that are being recognized or the results of studying a finite sample set of objects that are understandable in a certain sense. Each object is regarded as a chain or sentence, as it is composed of elements of the main vocabulary. The presented grammars can be used to classify objects. A presented unknown object can be attributed to a certain class if it is a language sentence. Otherwise, the object is assigned to another class. As a rule, an object is credited to the class in whose language it turns out to be a grammatically correct sentence. If this condition is not met, then it is obvious that the given object does not belong to any of the specified classes and, therefore, one more grammar is required, etc. The recognized object belongs to the class of interest if and only if it is a grammatically correct language sentence. If an object turns out to be a grammatically correct sentence of more than one language, the decision regarding its belonging is made in exactly the same way as it is done with other recognition methods, when it turns out that the results of the implementation of the recognition procedure do not allow the object to be definitely assigned to one of the specified classes.

The ratio of the separating force can be used as a criterion for the optimality of grammar. Words can be combined into more complex structures, that is, sentences. Here, language is seen as a set of sentences. Sentences are built from words and simpler sentences according to the syntax rules. The syntax of the language is a description of the correct sentences. The alphabet, vocabulary, and syntax completely define the set of allowed language constructs and the internal relationships between the constructs. A set of syntax rules form the grammar of a language. Syntax rules can describe either the procedure for obtaining correct sentences, or the procedure for recognizing the "correctness" of sentences (that is, their belonging to a given language). In the first case, the grammar will be generative, in the second it will be recognizing. The method of selection of the criterion of object recognition can be the simplicity of determining the measure of proximity, the complexity of writing the boundaries of classes and images, resolution, etc. In this case, it is very important to know the individual physical characteristics of the objects that are being recognized, the information content of the selected features, the quantity and quality of a priori and current information, the possibility of introducing adaptive (weighted) coefficients, etc.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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